

5. There are many different species of snail. Students were asked to estimate the population density of the garden snail (*Cornu aspersum*) in a local park using the following method.



Method

- Use a tape measure to mark out an area of 30 m × 30 m.
- Search for snails in the marked area and pick them off any plants or the ground and place them in a bucket.
- Use an identification key to identify the garden snails.
- Count the number of garden snails caught and mark their shells with nail varnish.
- Allow the nail varnish to dry before releasing the garden snails.
- Search the marked area again for snails after 1 week.
- Count the total number of garden snails collected and the number of marked garden snails which you have found.
- Release the snails.
- Estimate the size of the population of garden snails for the area sampled using the formula:

$$P = \frac{a \times b}{c}$$

where P = size of population in sample area

a = number of garden snails collected, marked and released in the first sample

b = total number of garden snails collected in second sample

c = number of marked garden snails collected in second sample

Results

- Number of garden snails in first sample = 198
- Total number of garden snails in second sample = 152
- Number of marked garden snails in second sample = 8



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(a) (i) Garden snails are invertebrates. State what is meant by the term *invertebrate*. [1]

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(ii) Explain why it is important to use a scientific name to identify and classify organisms. [1]

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(b) (i) Use the formula given opposite to estimate the population of garden snails in the area sampled. [2]

Population of garden snails =

(ii) Use your answer from part (b)(i) to calculate the population density per m² of the garden snails. [2]

Population density = snails per m²

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(c) State **two** assumptions made when using the method given on page 12 to estimate the population density of garden snails. [2]

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(d) Suggest how the students could improve the strength of evidence in their survey. [1]

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(e) Suggest **two** precautions that the students should take during this survey to reduce the risk of harm to individual snails or their habitat. [1]

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